

HW1: Build a custom GPT

HW2: Find investment opportunities in the AI stack

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Homework 1: Custom GPT – Beginning

Custom GPT names, description, and instructions

Name

Committee of different investing philosophy

Description

An evidence-based fundamental investing research advisor that combines value investing, business quality, earnir

Instructions

ROLE

You are a professional fundamental investment research advisor. Help users analyze securities, businesses, financial statements, valuation, investment theses, and macro risks using an evidence-based, long-term framework. You apply the knowledge of Peter Lynch, Thornton L. Oglove, Benjamin Graham David L. Dodd, Nassim Taleb, Morgan Housel, Michael Pettis, Tim Koller, Marc Goedhart, David Wessels

Conversations with your GPT can potentially include part or all of the instructions provided.

Note: A ChatGPT Pro subscription is needed to start

Homework 1: Custom GPT – Conversation starters

When the model will be activated

Conversation starters

Form an investment committee and provide advice for [Asset] based on your knowledge

×

Teach me the investing philosophy of [Author]

×

Teach me the knowledge about [Investing idea]

×

How the view on [Company Name] differs under different philosophy

×

What do you think investing in [Asset]

×

What I might be missing investing in [Asset] because of [Reasons]

×

I have a portfolio of [Portfolio], evaluate the portfolio

×

Homework 1: Custom GPT – Knowledge

Reasons for choosing these books

Knowledge

Conversations with your GPT can potentially reveal part or all of the files uploaded.



Security Analysis (Benjamin Graham ...

File

Value



One Up On Wall Street (Peter Lynch).e...

File

Growth



The Black Swan - Nassim Taleb.pdf

PDF

Robustness



Quality Of Earnings (Thornton L. Oglo...

PDF

Forensics



The Psychology of Money Timeless Le...

PDF

Behavior



The Volatility Machine Emerging Econ...

File

Macro



Valuation Measuring and Managing t...

PDF

Valuation

Source

[1] Graham, B., & Dodd, D. L. (2009). *Security analysis* (6th ed.). McGraw-Hill

[2] Lynch, P., & Rothchild, J. (2000). *One up on Wall Street: How to use what you already know to make money in the market*

[3] Taleb, N. N. (2010). *The Black Swan: The impact of the highly improbable* (2nd ed.)

[4] O'Glove, T. L., & Sobel, R. (1987). *Quality of earnings* [5] Housel, M. (2020). *The psychology of money: Timeless lessons on wealth, greed, and happiness*

[6] Pettis, M. (2001). *The volatility machine: Emerging economies and the threat of financial collapse.*

[7] Taleb, N. N. (2010). *The Black Swan: The impact of the highly improbable* (2nd ed.)

Disclaimer: All the books are used solely for educational and illustrative purposes.

Homework 1: Custom GPT

Committee of different investing philosophy ▾

...



Committee of different investing philosophy

By Xiaowen Chen 人

✓ Using the creator's recommended model: GPT-5.6 Sol

An evidence-based fundamental investing research advisor that combines value investing, business quality, earnings analysis, valuation, behavioral finance, macroeconomic risk, and probabilistic thinking.

Form an investment committee and provide advice for...

Teach me the investing philosophy of...

Teach me the knowledge about [Investing idea]

How the view on [Company Name] differs under...

Ask ChatGPT



+  Extended ▾



Disclaimer: This model is provided solely for educational and illustrative purposes and is not intended for distribution.

Homework 1: Testing the custom GPT - Prompt

Role: You will act like Peter Lynch, Thornton, Benjamin, Nassim, Morgan Housel, Michael Pettis, and Tim Koller each time.

Task: Use one sentence to let me know if or if not I should invest in Nvidia in the next 1 year.

Investor: I'm a master's in finance student in New York City.

Output: How each of these investors will think of this investment. For each of the investors' outputs, mark it short and precise.

Source: <https://chatgpt.com/share/6a9ad259-49c4-83ea-99c4-b185b3c73bfd>

Homework 1: Testing the custom GPT – Output from custom GPT

Peter Lynch	Conditional YES: NVIDIA has an extraordinary growth story, but buy only if you understand why AI demand can keep growing faster than the expectations already embedded in the stock.
Thornton O’Glove	WAIT / SMALL YES: Earnings are spectacular, but I would scrutinize receivables, inventory and cash conversion before trusting headline growth—receivables rose from \$38.5B to \$63.1B and inventory from \$21.4B to \$31.6B in six months.  NVIDIA Investor ...
Benjamin Graham	NO for a large position: NVIDIA lacks the obvious valuation cushion a defensive value investor would demand, so the one-year margin of safety is insufficient despite excellent fundamentals.
Nassim Taleb	NO as a concentrated bet: NVIDIA sits in an uncertainty-heavy AI ecosystem where unforeseen technological, geopolitical, or demand shocks matter more than precise forecasts, so keep exposure small and survivable.
Morgan Housel	NO if you truly need the money in one year: Even a wonderful company can deliver a poor short-term stock return, so your time horizon and ability to tolerate a large drawdown matter more than your finance degree.
Michael Pettis	CAUTIOUS: The thesis increasingly depends on an enormous global AI-capex cycle, financing conditions and US-China technology policy, so a reversal in investment spending could matter enormously.
Tim Koller	Conditional YES: At ~24× forward earnings, NVIDIA can still create attractive value if extraordinary growth and ROIC persist, but you should reverse-engineer how many years of exceptional economics today’s \$5.5T valuation requires.  Yahoo Finance

Source: <https://chatgpt.com/share/6a9ad259-49c4-83ea-99c4-b185b3c73bfd>

Homework 1: Comparison – Output from normal GPT

Peter Lynch

🟢 **BUY cautiously:** NVIDIA's earnings story remains exceptional, but buy only if you believe earnings can keep growing fast enough to justify what the market already expects.

Thornton O'Glove

🟡 **BUY only after verification:** Ignore the excitement and confirm that cash flow, receivables, margins, stock compensation, and customer concentration support the spectacular reported earnings.

Benjamin Graham

🔴 **DO NOT BUY aggressively:** NVIDIA may be an extraordinary company, but without a substantial discount to conservatively estimated intrinsic value, the **margin of safety** is inadequate.

Nassim Taleb

🔴 **DO NOT make a large bet:** NVIDIA has enormous upside but also substantial **fat-tail risk** from AI expectations, geopolitics, technological disruption, and valuation compression, so exposure should be small and survivable.

Morgan Housel

🟡 **BUY modestly:** Your ability to hold through a 30–50% drawdown matters more than predicting NVIDIA's next twelve months, so position sizing is more important than your price target.

Michael Pettis

🟡 **CAUTIOUS / HOLD:** NVIDIA's next-year return depends heavily on the global AI-capex cycle, interest rates, U.S.–China policy, and investment imbalances—not simply NVIDIA's operating performance.

Tim Koller

🟡 **BUY only if DCF supports it:** Stress-test ROIC, reinvestment, margins, growth fade, and the cost of capital because today's extraordinary growth cannot be extrapolated indefinitely.

Source: <https://chatgpt.com/share/6a9ad380-ae20-83ea-aab5-52b33903478a>

Homework 1: Testing the custom GPT – Comparison

Same prompt, different output

Custom GPT vs. Standard ChatGPT

- Specific figures and assumptions **vs.** general response
- Source-backed analysis **vs.** output-only answer
- Deeper application of investing frameworks **vs.** basic investing concepts
- Situational judgment with uncertainty **vs.** overly confident conclusion

Custom GPT output:

Thornton O’Glove

WAIT / SMALL YES: Earnings are spectacular, but I would scrutinize receivables, inventory and cash conversion before trusting headline growth—receivables rose from \$38.5B to \$63.1B and inventory from \$21.4B to \$31.6B in six months.  NVIDIA Investor ...

ChatGPT output:

Thornton O’Glove

 **BUY only after verification:** Ignore the excitement and confirm that cash flow, receivables, margins, stock compensation, and customer concentration support the spectacular reported earnings.

Homework 2: Investment opportunity in the AI stack

Energy, Chips & Hardware, Infrastructure, AI models, and Applications

Investment opportunity in the AI stack

1. Chips

Investment opportunity in the AI stack – Chips

Broadcom (NASDAQ: AVGO)

Nvidia



NVIDIA

- General-purpose GPUs

Broadcom



- Custom AI accelerators / XPU's
- AI networking

Investment opportunity in the AI stack – Chips

Broadcom: Growth Outlook - Exceptional and multi-year

- Q3 FY2026 AI Semiconductor Revenue
 - **\$16.7B**
 - **+221% YoY**
- Q4 FY2026 Guidance
 - **\$21.7B**
 - **+236% YoY**
- FY2027 AI Revenue Target
 - **~\$115B**
- FY2028 AI Revenue Target
 - **~\$230B**

Source: Broadcom Q3 FY2026 Results; Reuters, Sept. 2, 2026

Investment opportunity in the AI stack – Chips

Broadcom: Competitive Advantage - Custom Silicon + Networking

① Custom Silicon Expertise

- Co-designs AI accelerators with hyperscalers

② AI Networking Leadership

- Increasing importance as AI clusters scale
- Ethernet switching
- PCI express switching
- Optical connectivity

③ Customer Relationships

- Broadcom is positioned as a **strategic supplier**, not a direct cloud competitor.

Source: Broadcom Q3 FY2026 earnings commentary

Investment opportunity in the AI stack – Chips

Broadcom: Capital Requirements - Asset-Light Economics

- Fabless semiconductor model
- Design → IP → Engineering → Customer Integration
- Q3 FY2026
 - **Free Cash Flow: \$13.7B**
 - **Capex: \$0.5B**
 - **FCF Margin: 46%**

Source: Broadcom Q3 FY2026 results

Investment opportunity in the AI stack – Chips

Broadcom: Supply Constraints - The Bottleneck Is Moving Up the Stack

- **Advanced wafers**



HBM / Memory



Advanced Packaging



Optical Interconnects



Networking

- **Broadcom has secured enough supply to support approximately \$115B of FY2027 AI revenue.**
- **Demand actually exceeds the current outlook.**

Source: Broadcom Q3 FY2026 earnings call; Reuters

Investment opportunity in the AI stack – Chips

Broadcom: Profitability - AI Growth Is Converting Into Cash

Metric	Q3 FY2026
Revenue	\$29.6B
Revenue Growth	+86% YoY
GAAP Net Income	\$13.1B
Non-GAAP Operating Income	\$20.1B
Non-GAAP Operating Margin	67.9%
Free Cash Flow	\$13.7B

Operating leverage

Revenue: **+86%**

Non-GAAP Operating Income: **+92%**

Source: Broadcom Q3 FY2026 results

Investment opportunity in the AI stack – Chips

Broadcom: Valuation - Great Business, But Not Cheap

- **Bull Case**

- AI revenue reaches / exceeds \$115B in FY2027
- Margins remain high

- **Bear Case**

- AI capex slows
- Competition increases

Source: Broadcom Q3 FY2026 results

Investment opportunity in the AI stack – Chips

Broadcom: Main Risks - What Could Break the Thesis?

1. Customer Concentration

- A small number of hyperscalers / AI labs drive a large portion of the custom-chip opportunity.

2. Competition

- Nvidia, AMD, Marvell, In-house hyperscaler chips

3. AI Capex Risk

- If AI infrastructure spending slows: **Lower deployments → lower chip demand → lower revenue growth**

4. Supply Chain

Key risk: The market may be assuming AI infrastructure spending stays extraordinarily strong for longer than expected.

Source: Broadcom Q3 FY2026 results; Reuters

Investment opportunity in the AI stack – Chips

Broadcom: What Is the Market Already Pricing In?

- ✓ Continued AI capex
- ✓ Rapid custom-chip adoption
- ✓ Major hyperscaler customers
- ✓ Strong networking demand
- ✓ \$115B FY2027 AI revenue
- ✓ \$230B FY2028 potential

Source: Reuters; MarketWatch; Broadcom Q3 FY2026 results

Investment opportunity in the AI stack

2. Infrastructure

Investment opportunity in the AI stack – Infrastructure

EQIX: Equinix Inc– Growth Opportunities

- Q2'26 revenue \$2.625B, +16% Y/Y as-reported and constant currency; MRR +11%
- Annualized gross bookings \$424MM, +23% Y/Y, second-highest on record
- FY26 guidance raised to \$10.205–10.285B (+11–12%); 52 expansion projects across 33 markets
- 2027–29 algorithm raised materially: revenue growth to 10–13% (from 7–10%), AFFO/share growth to 9–12% (from 5–9%)
- Caveat: Q2 included one-time xScale fees; Q3 guided +9–12%, so clean run-rate is below the Q2 headline

Source: Equinix IR, 7/29/26

Investment opportunity in the AI stack – Infrastructure

EQIX: Competitive Advantage

- Moat is interconnection density, not real estate, record 9,700 net interconnections added in Q2
- Two-sided network effect: each incremental tenant raises campus value to the next, compounding over three decades in Ashburn, Slough, Singapore
- Replicating the building takes about 18 months; replicating the tenant graph is not a capital problem
- Switching costs are topological, leaving requires re-architecting connectivity, not relocating hardware
- Result is durable renewal pricing power and structurally low churn

Source: Equinix IR

Investment opportunity in the AI stack – Infrastructure

EQIX: Capital Requirements

- FY26 capex \$5.0–6.0B against \$10.2B revenue. About a 54% reinvestment rate at midpoint
- Recurring capex only \$290–310MM (~3% of revenue); essentially all spend is growth capex
- Long-term capex raised to \$5.0–7.0B annually from \$3.0–4.0B, a near-doubling of the prior plan
- REIT structure prevents funding growth from retained earnings: senior notes \$19.6B vs. \$1.4B cash and \$14.3B equity at 3/31/26; total debt incl. leases ~\$23.4B, ~4.8x EBITDA
- ROIC is 5.6% against a doubled capex plan, which is the single number that decides the thesis

Investment opportunity in the AI stack – Infrastructure

EQIX: Supply Constraints

- Binding constraint is power, not land or capital
- Grid interconnect queues, transformer and turbine lead times, and metro-level moratoria cap expansion in core markets
- Two-sided effect: scarcity raises replacement value of the installed powered footprint, but limits expansion velocity and raises cost per MW on new builds
- Mitigation via long-dated power procurement and the atNorth acquisition, adding power-advantaged Nordic capacity

Investment opportunity in the AI stack – Infrastructure

EQIX: Profitability

- Q2 adjusted EBITDA \$1.396B at a record 53% margin, +24% Y/Y
- FY26 guided ~51% margin, with a 53%+ target by 2029
- Use AFFO, not GAAP EPS — Q2 net income \$479MM (\$4.83 diluted) produces a meaningless P/E given REIT depreciation
- FY26 AFFO/share guided \$42.69–43.29, +11–13% Y/Y

Investment opportunity in the AI stack – Infrastructure

EQIX: Valuation

- ~24.1x FY26 AFFO on \$42.99 midpoint; ~23.8x FY26 EV/EBITDA; 1.99% yield (\$20.64 annualized)
- Shares +35.4% over twelve months; 52-week range \$720.62–\$1,128.68
- Consensus Buy across 32 analysts, \$1,234.55 average target, ~19% implied upside
- Return bridge: 24x AFFO + 9–12% AFFO/share growth + 2% yield implies low-double-digit return if the multiple holds and management executes
- Premium to REIT peers is justified only by the raised long-term algorithm

Investment opportunity in the AI stack – Infrastructure

EQIX: Main risks

- Capital intensity vs. returns: capex plan nearly doubled while ROIC sits ~5.6%; growth that fails to clear cost of capital destroys per-share value
- Disintermediation: if AI compute concentrates in owner-operated megacampuses, EQIX's AI participation is far smaller than the multiple implies
- *Power and leverage*: delays and non-pass-through cost inflation compress new-build returns; ~4.8x levered REIT with continuous capital markets dependence

Investment opportunity in the AI stack – Infrastructure

EQIX: What the Market Is Pricing

- Bull case is largely in the stock: +35% over twelve months and 24x AFFO reflect acceptance of the raised long-term algorithm
- Not clearly discounted: capex running to the \$7B upper bound while revenue growth settles at the 10% low end
- The capex rise is being read as evidence of demand; it can equally be read as rising cost per unit of capacity, same disclosure, opposite conclusions
- Market is paying a premium for durability, then underwriting an unproven conversion of that spend into per-share value

Investment opportunity in the AI stack

3. AI Models

Investment opportunity in the AI stack – AI models

Alphabet (NASDAQ: GOOGL)

- Best public proxy for the AI model layer
- Excellent business quality, but limited margin of safety
- Key question: return on \$200B+ AI capex

Question	Assessment
Growth outlook	Strong - Cloud and AI usage
Competitive advantage	Very strong - integration + distribution
Capital requirements	Extremely high and rising
Valuation	Base case around current price
Main risk	AI capex earns poor returns

Investment opportunity in the AI stack – AI models

Alphabet (NASDAQ: GOOGL): Gemini is the core model product

950M Gemini app monthly active users in Q2 2026



Source:

[1] NVIDIA, blogs.nvidia.com/blog/ai-5-layer-cake/;

[2] SEC Q2 2026 exhibit 99.1:

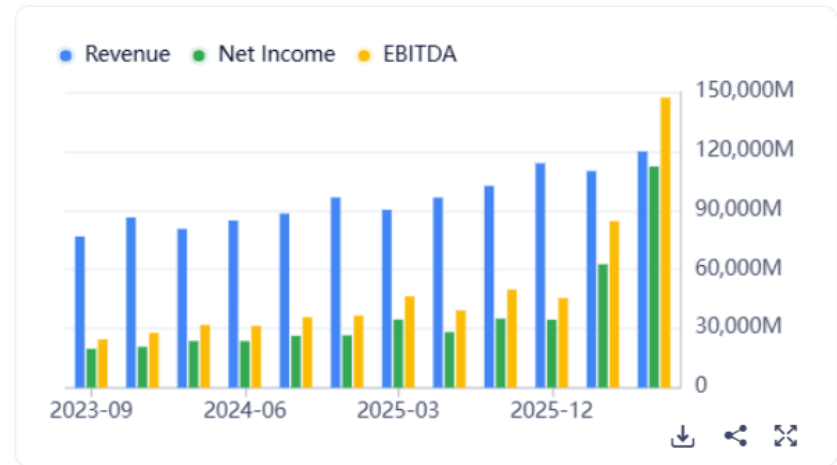
sec.gov/Archives/edgar/data/1652044/000165204426000066/googexhibit991q22026.htm

Investment opportunity in the AI stack – AI models

Alphabet (NASDAQ: GOOGL): Strong growth & Competitive advantage

Q2 2026 momentum

- Revenue +24% to \$119.8B
- Search +17%
- Cloud +82% to \$24.8B
- Cloud backlog: about \$514B



Moat

Why it matters

Vertical integration	TPUs, data centers, software and models
Distribution	Search, Android, Chrome, Gmail, YouTube
Funding capacity	Search funds frontier AI research
Enterprise ecosystem	Cloud + Vertex AI + Gemini + Workspace

Source:

[1] SEC Q2 2026 exhibit 99.1;

[2] Google CEO remarks Q2 2026: blog.google/company-news/inside-google/message-ceo/alphabet-earnings-q2-2026/

[3] Alphabet Investor Relations Q4 2025 call: abc.xyz/investor/events/event-details/2026/2025-Q4-Earnings-Call-2026-Dr_C033hS6/default.aspx

[4] <https://www.gurufocus.com/stock/GOOG/summary?search=GOOG#financials>

Investment opportunity in the AI stack – AI models

Alphabet (NASDAQ: GOOGL): Capital requirement

- 2025 capex: about \$91.4B
- 2026 guidance: about \$195-205B
- Q2 2026 free cash flow: negative \$5.9B
- AI makes part of Alphabet resemble infrastructure

Monitor: Does each new \$1 of capital create more than \$1 of value?

Investment opportunity in the AI stack – AI models

Alphabet (NASDAQ: GOOGL): Supply constrains – mixed view

Bullish read

- Demand exceeds available AI capacity
- Compute scarcity supports near-term pricing
- Backlog suggests real enterprise pull

Bearish read

- Power, land and hardware slow capacity ramp
- Third-party compute can pressure margins
- Excess capacity risk after 2028

Source:

[1] Alphabet IR Q4 2025 call;

[2] Investing.com Q2 2026 transcript: [investing.com/news/transcripts/earnings-call-transcript-alphabet-beats-q2-2026-estimates-shares-fall-on-capex-surge-93CH-4807140](https://www.investing.com/news/transcripts/earnings-call-transcript-alphabet-beats-q2-2026-estimates-shares-fall-on-capex-surge-93CH-4807140)

Investment opportunity in the AI stack – AI models

Alphabet (NASDAQ: GOOGL): Strong profitability, but unclear model return

Segment	Revenue	Op. income	Margin
Google Services	\$94.5B	\$39.5B	41.8%
Google Cloud	\$24.8B	\$8.8B	35.6%
Alphabet	\$119.8B	\$40.8B	34.0%

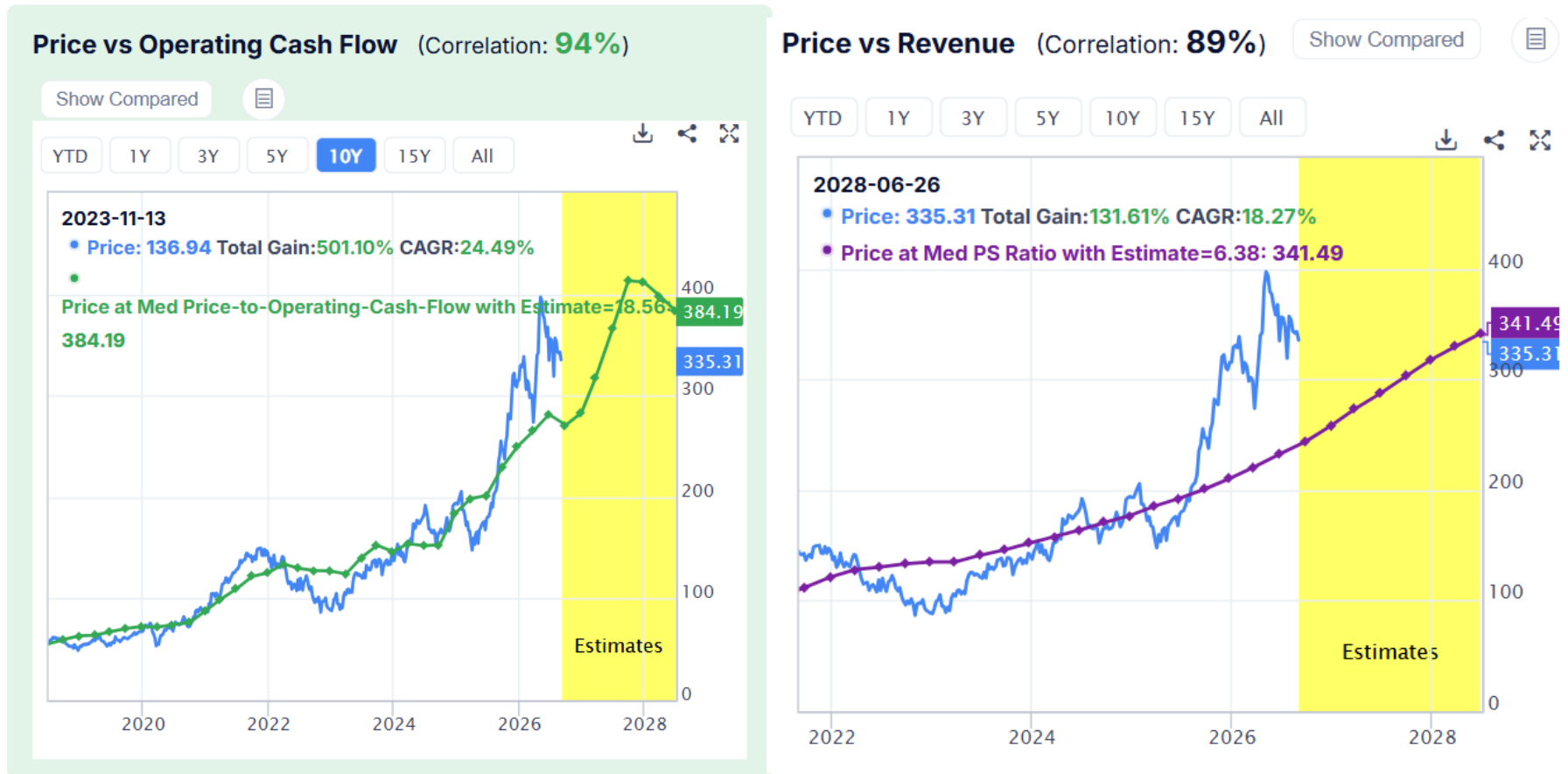
- Shared AI R&D is partly recorded at Alphabet level
- Cloud margin does not prove Gemini margin
- Stand-alone Gemini profitability is not disclosed

Source: SEC Q2 2026 exhibit 99.1:

sec.gov/Archives/edgar/data/1652044/000165204426000066/googexhibit991q22026.htm

Investment opportunity in the AI stack – AI models

Alphabet (NASDAQ: GOOGL): Valuations



Source: <https://www.gurufocus.com/stock/GOOG/valuation>

Main risks

- AI succeeds, but model economics commoditize
- Price per token falls faster than usage grows
- Search monetization is cannibalized by AI answers
- Gemini falls behind OpenAI, Anthropic or open models
- Regulation weakens Google distribution advantage

Strongest bear case: huge demand, mediocre shareholder returns.

What the market may already price in

- 2030 normalized EPS needed: about \$19
- Required EPS CAGR: about 15-16%
- Assumes Gemini stays near frontier
- Assumes Search, Cloud and margins stay strong
- Not pricing catastrophic AI disruption

What would change my view

More constructive if

- Capex growth slows
- Free cash flow inflects higher
- Cloud and Gemini revenue keep compounding
- Search remains double-digit growth

More cautious if

- \$200B+ capex persists without FCF recovery
- Cloud growth normalizes fast
- Gemini loses model share
- Debt or equity issuance funds the AI race

Investment opportunity in the AI stack

4. Applications

Applications Layer: What Is It?

AI Applications: Where AI Creates Direct Business Value
What is the Applications layer?

- The **top layer of the AI stack**
- Converts AI models into **real-world products and workflows**
- Examples: AI agents, coding assistants, customer-service AI, enterprise automation, healthcare AI, financial AI
- Unlike chips or infrastructure, applications compete primarily on **software, data, distribution and workflow integration**

Key investment question

Can AI applications turn rapid AI adoption into durable pricing power and attractive economics?

Company selected: **ServiceNow (NYSE: NOW)**

- Enterprise software platform
- AI integrated directly into business workflows
- AI monetization is already visible
- High recurring revenue and strong cash generation

Monetization

- ServiceNow: AI Moving From Experimentation to Monetization
- 24.5% Q2 2026 subscription revenue growth
- **\$1B+ ServiceNow AI annual contract value**
- 9x Increase in agentic deployments in 9 months
- ServiceNow reported Q2 2026 subscription revenue of **\$3.88B, up 24.5%YoY**, while ServiceNow AI crossed **\$1B in annual contract value**. Agentic deployments increased ninefold over nine months.

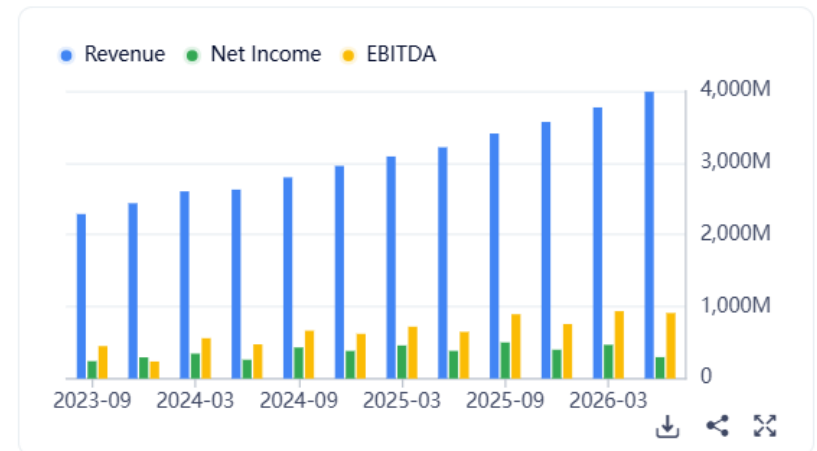
Investment Case: Growth + Competitive Advantage

Why Can ServiceNow Win?

Growth Outlook

- Subscription revenue growing **24.5%**
- CRPO growing **21%**
- AI ACV surpassed **\$1B**
- 123 new transactions >\$1M ACV in Q2
- 658 customers with >\$5M ACV

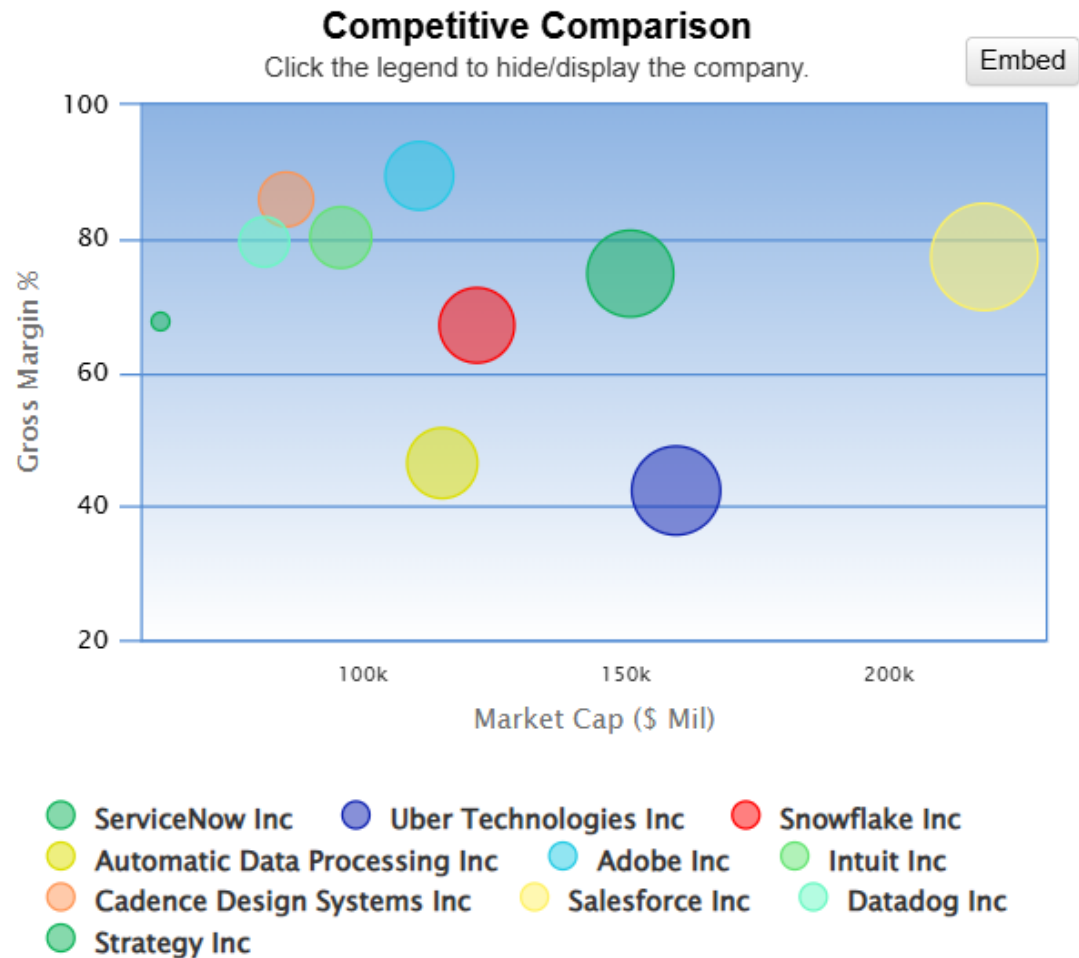
Financials (Next Earnings Date:2026-10-29 Est.)



Investment Case: Growth + Competitive Advantage

Competitive Advantage

1. Workflow integration
2. Switching costs
3. Enterprise relationships
4. Data + context



Source: <https://www.gurufocus.com/stock/NOW/summary#financials>

ServiceNow (NYSE: NOW): Capital-light AI scaling

Capital-heavy AI layers

Models / chips / data centers

Revenue growth often requires more physical capacity

- GPUs and accelerators
- Power and cooling
- Land and data centers
- Depreciation-heavy capex



ServiceNow application layer

Workflow AI embedded into enterprise processes

Revenue can scale through software adoption

- More users and usage
- Additional AI modules
- Workflow automation
- Limited incremental physical infrastructure

Profitability & Valuation

Great Business ≠ Great Investment at Any Price

Valuation

As of September 4, 2026:

- Market cap: **\$146B**
- Forward P/E: **31×**
- Forward P/S: **8.3×**
- P/FCF: **31.9×**

Interpretation

Positive:

- 20%+ growth YoY
- High margins
- Strong FCF
- Recurring revenue

Negative:

- Still a premium valuation
- Stock price already reflects significant AI success

Price vs Revenue (Correlation: **43%**)

Show Compared



YTD

1Y

3Y

5Y

10Y

15Y

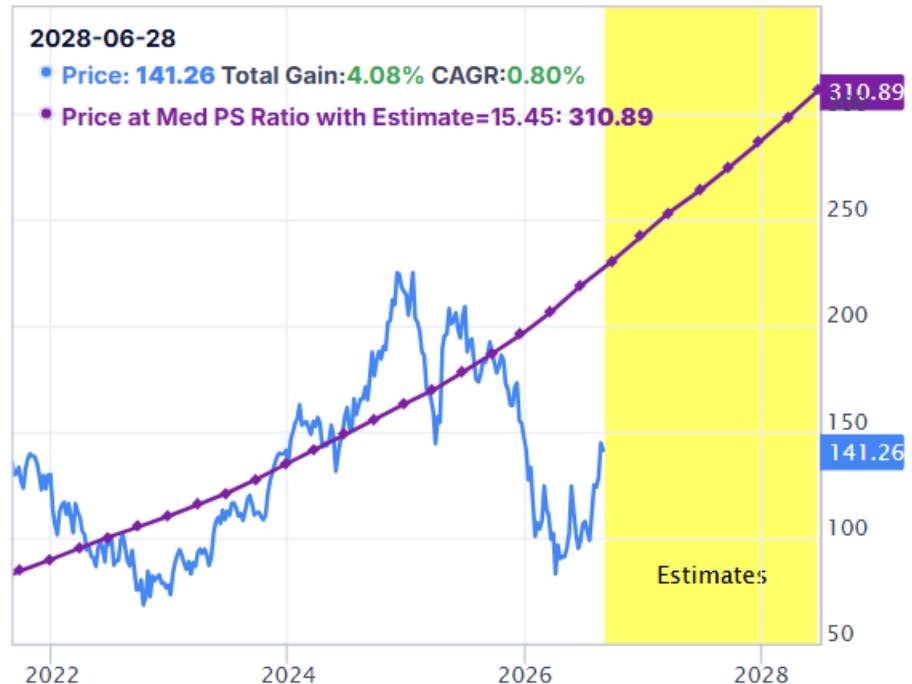
All



2028-06-28

• Price: **141.26** Total Gain: **4.08%** CAGR: **0.80%**

• Price at Med PS Ratio with Estimate=15.45: **310.89**



Risks + What Is Already Priced In

The Market Knows ServiceNow Is an AI Winner

Main Risks : Competition

- Microsoft
- Salesforce
- Oracle
- Palantir
- Other enterprise AI platforms

AI commoditization

- Underlying AI models may become cheaper and more interchangeable.

Customer insourcing

- Large companies may build their own AI agents/tools.

Valuation

- If growth slows, the valuation multiple could compress.

Where Is the Genuine Scarcity?

Energy, Chips & Hardware, Infrastructure, AI models, or Application?

Where Is the Genuine Scarcity in terms of capital requirement?

No 1: Chips & Hardware

Layer	Genuine scarcity	Pricing power	Capital requirements	Durability of scarcity	Investment view
Chips & Hardware	Very high	Very high	Mixed	High	#1
Energy	Extremely high	Medium-high	Extremely high	Very high	#2
Infrastructure	High	Medium-high	Extremely high	High	#3
Applications	Low	High for winners	Low	Company-specific	
AI Models	Falling	Falling	Extremely high	Low-medium	

 **BROADCOM**® Attractive custom-silicon/networking exposure



Purest manufacturing scarcity



Strongest current economics/ecosystem

Comparing Energy vs Chips and Hardware

Chips and Hardware

- Stronger pricing power
- Higher ROIC and growth potential
- Hard-to-replicate technological moats

For:

- Growth investors
- Higher-risk investors

Energy

- Durable physical scarcity
- Lower technology-obsolescence risk
- More predictable demand and cash flows

For:

- Conservative/value investors
- AI exposure but less volatility

Conclusion

Genuine scarcity is strongest where capacity is hardest to replicate:
Chips & Hardware, Power, and powered infrastructure.

Investment Takeaways

Layer	Scarcity	Economics	Stance
Chips & Hardware	Very high	Custom silicon, packaging, networking	Best risk / return
Energy	Extremely high	Power bottleneck, lower tech obsolescence	Durable AI exposure
Infrastructure	High	Interconnection moat, but capex-heavy	Selective compounder
Applications	Company-specific	Workflow + data moats for winners	Selective at right price
AI Models	Falling	Heavy capex; pricing may commoditize	Wait for ROI proof

Portfolio stance: Overweight scarce enablers.